

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

# Section 5 — Security Visual Maps

GCP Study Notes

Study Notes   April 2026

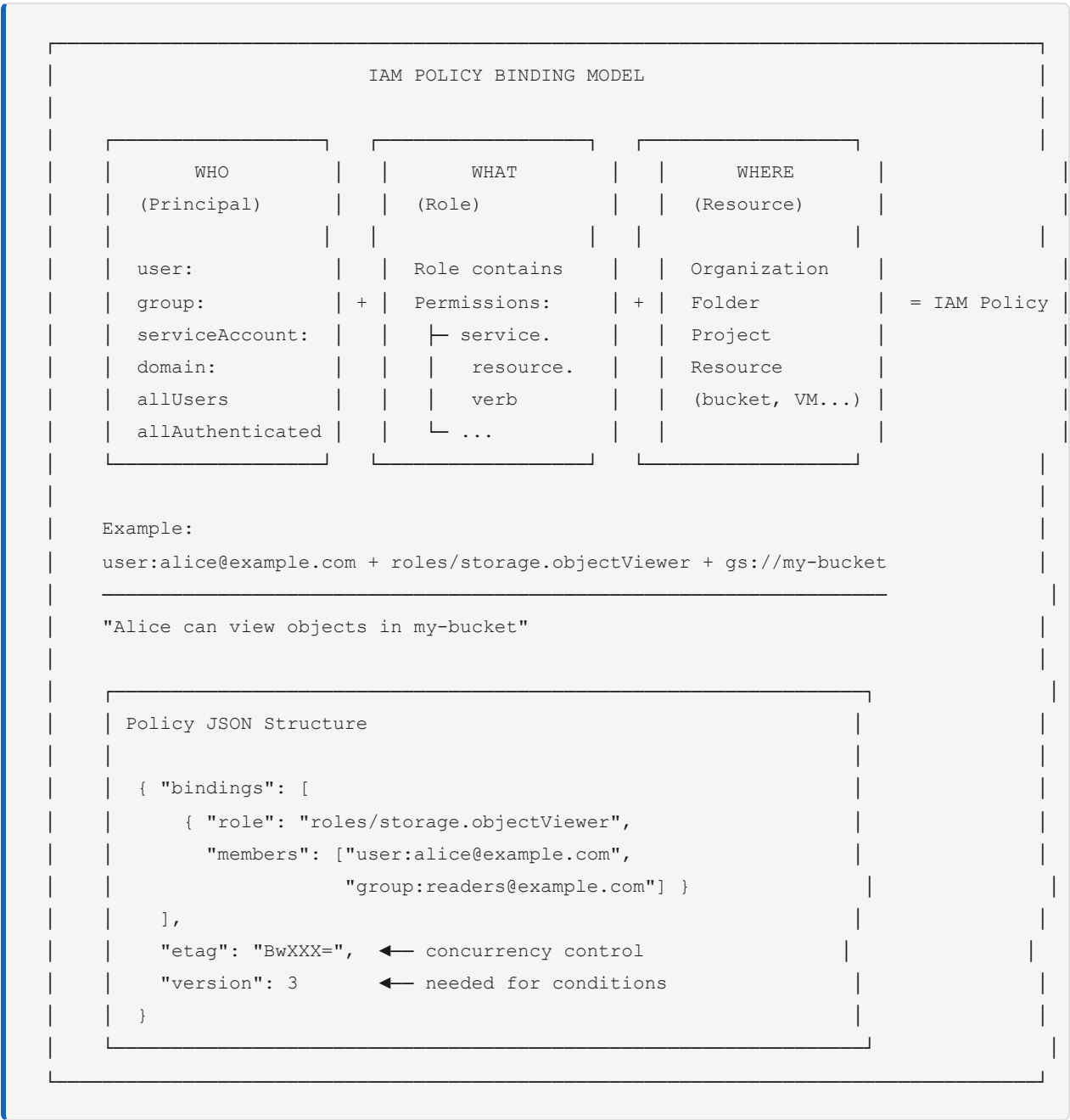
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## IAM & Service Accounts

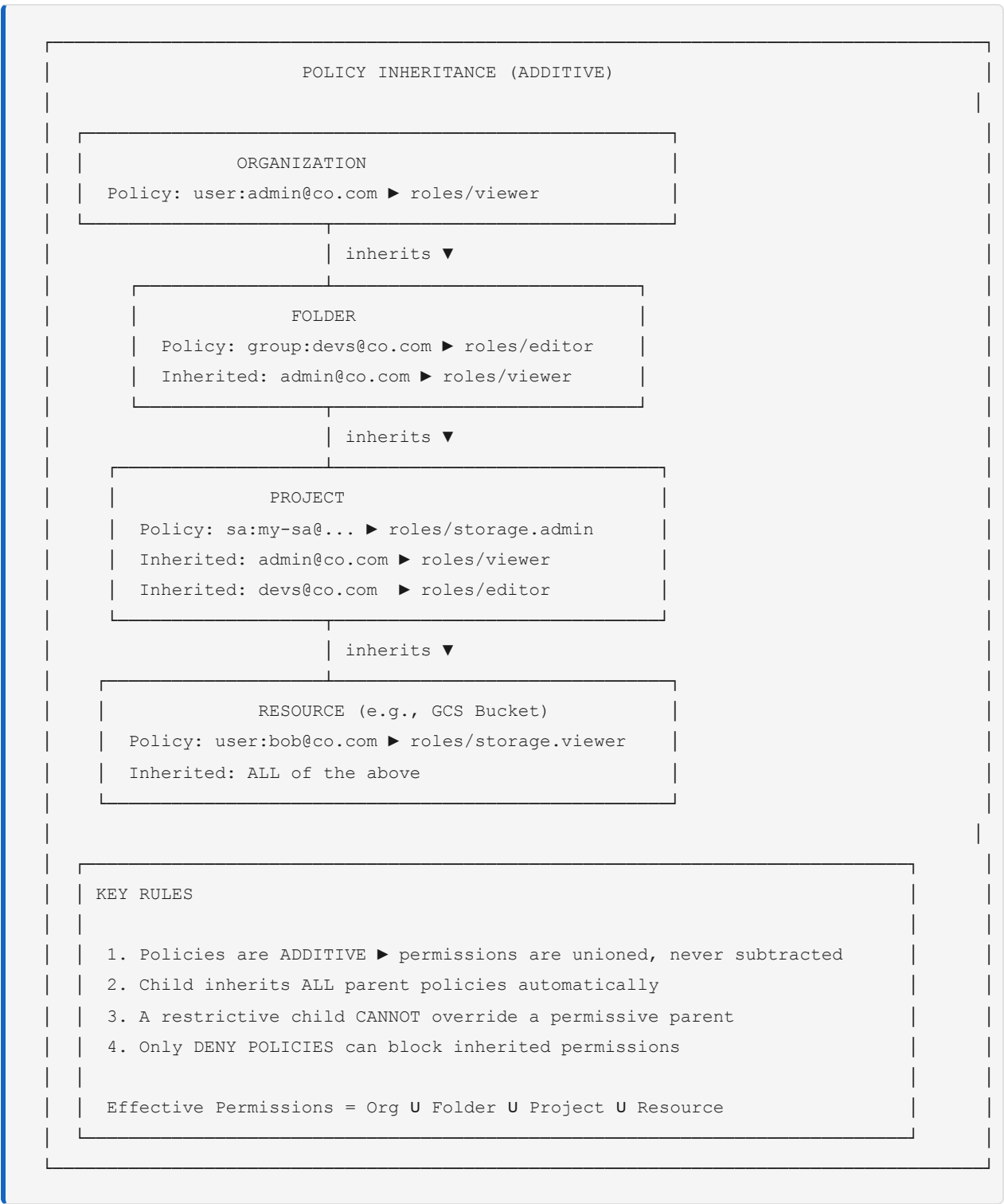
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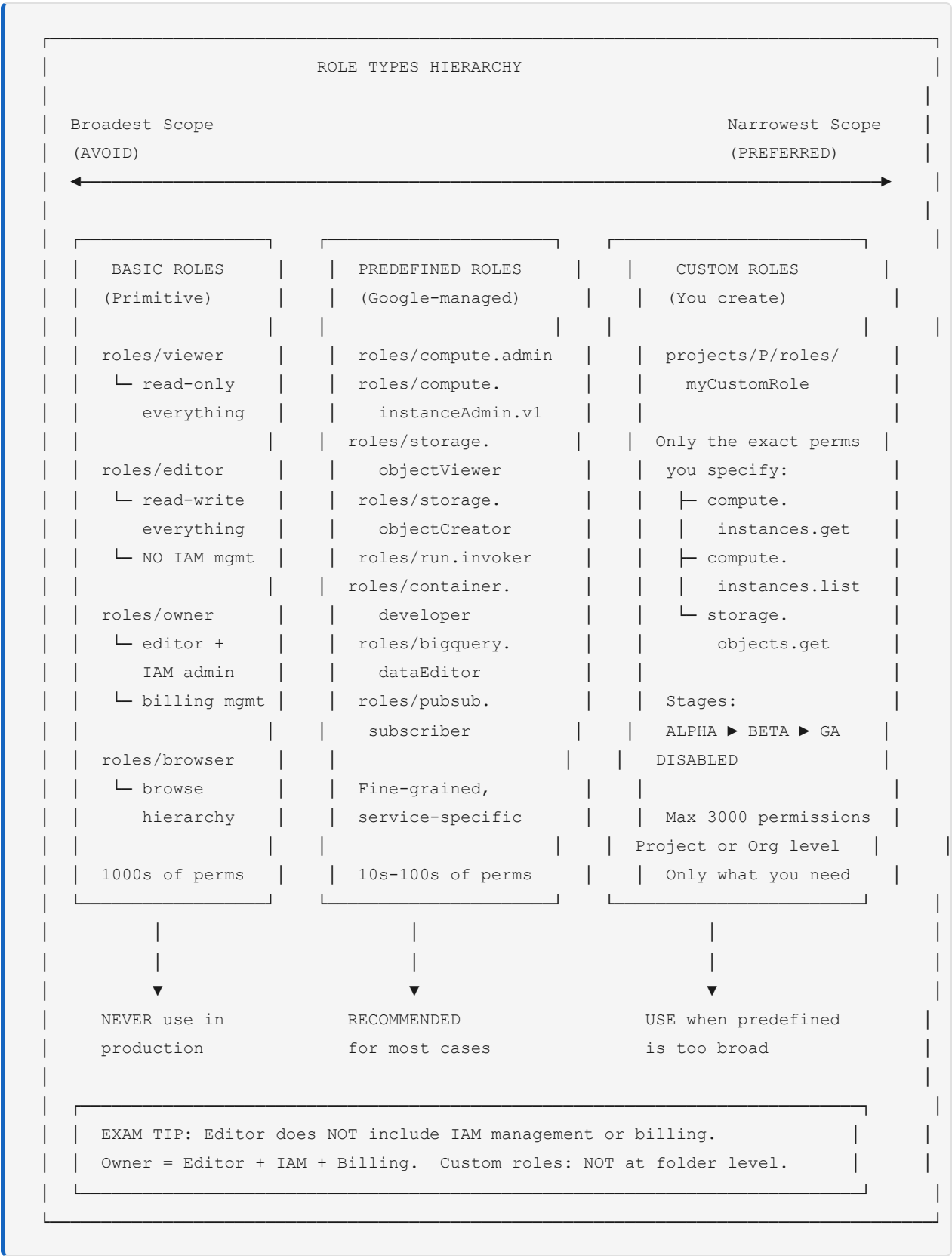
1. IAM Model Diagram: WHO + WHAT + WHERE = Policy



2. Policy Inheritance Flow: Additive Nature



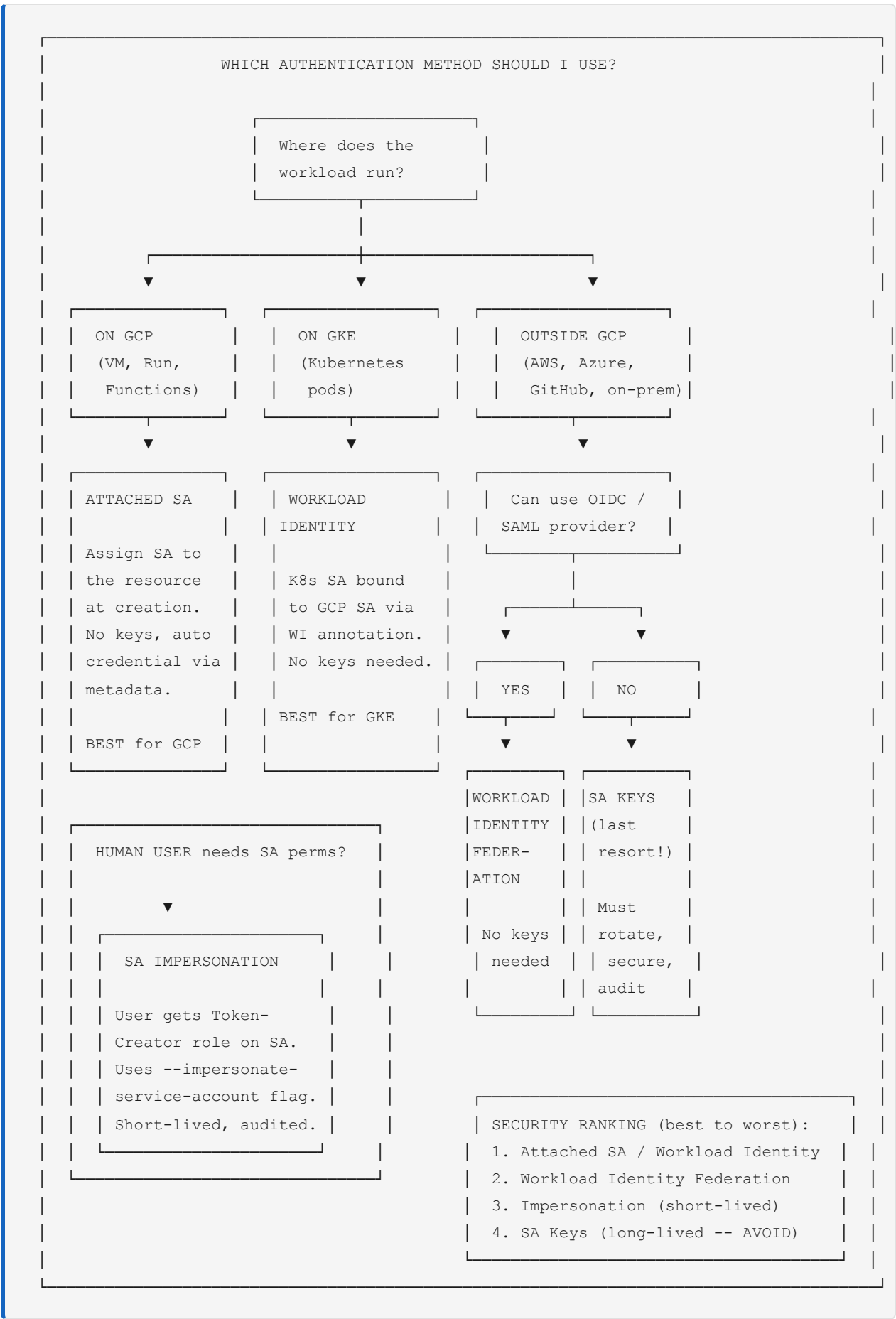
3. Role Types Hierarchy: Basic, Predefined, Custom



4. Service Account Identity Diagram: Principal vs Resource

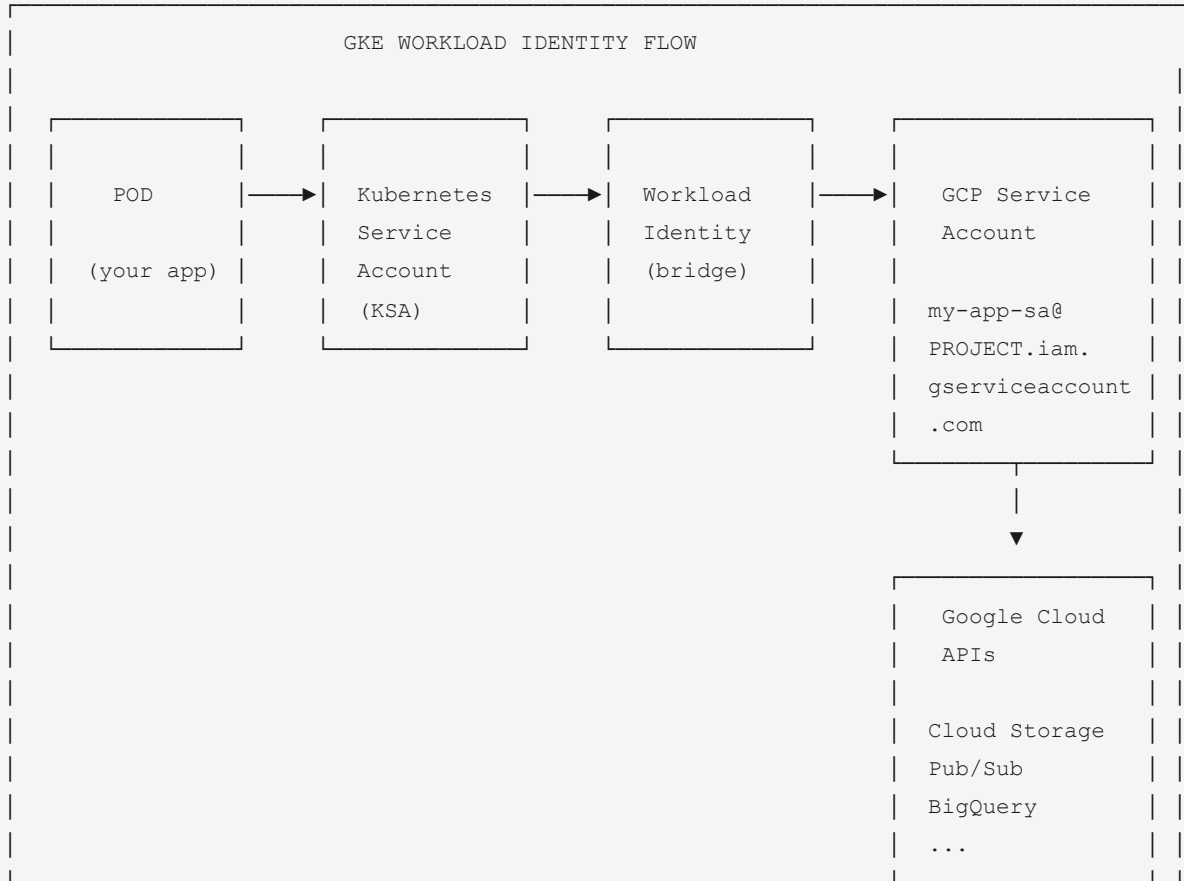


5. Authentication Methods Flowchart





## 6. Workload Identity Flow (GKE)



### SETUP STEPS:

#### Step 1: Annotate K8s Service Account

```
kubectl annotate serviceaccount my-ksa \
  --namespace=default \
  iam.gke.io/gcp-service-account=my-app-sa@PROJECT.iam.g...
```

#### Step 2: Allow KSA to impersonate GCP SA

```
gcloud iam service-accounts add-iam-policy-binding \
  my-app-sa@PROJECT.iam.gserviceaccount.com \
  --role="roles/iam.workloadIdentityUser" \
  --member="serviceAccount:PROJECT.svc.id.goog[NAMESPACE/KSA_NAME]"
```

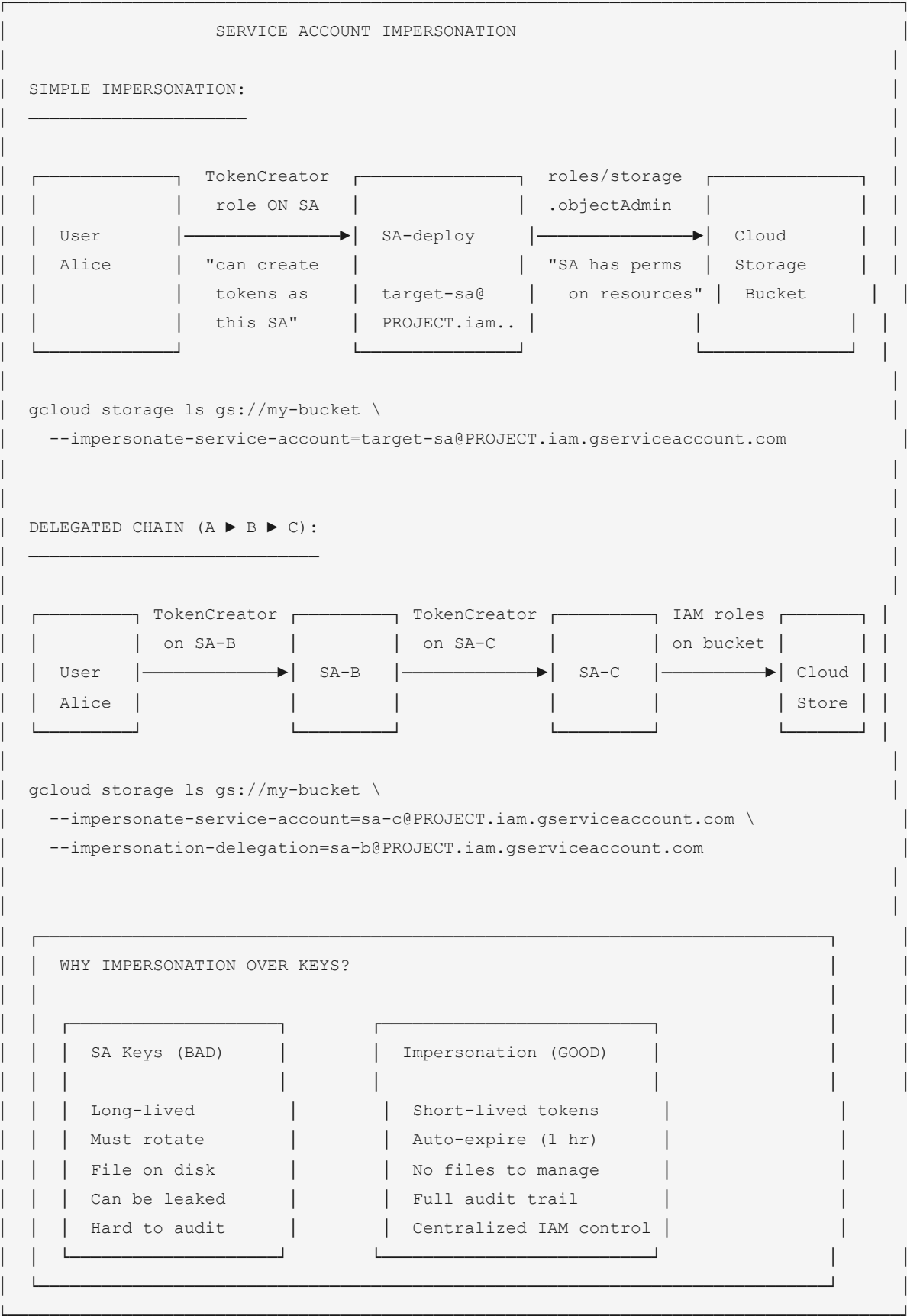
#### Step 3: GCP SA has the needed IAM roles on resources

```
gcloud projects add-iam-policy-binding PROJECT_ID \
  --member="serviceAccount:my-app-sa@PROJECT.iam.g..." \
  --role="roles/storage.objectViewer"
```

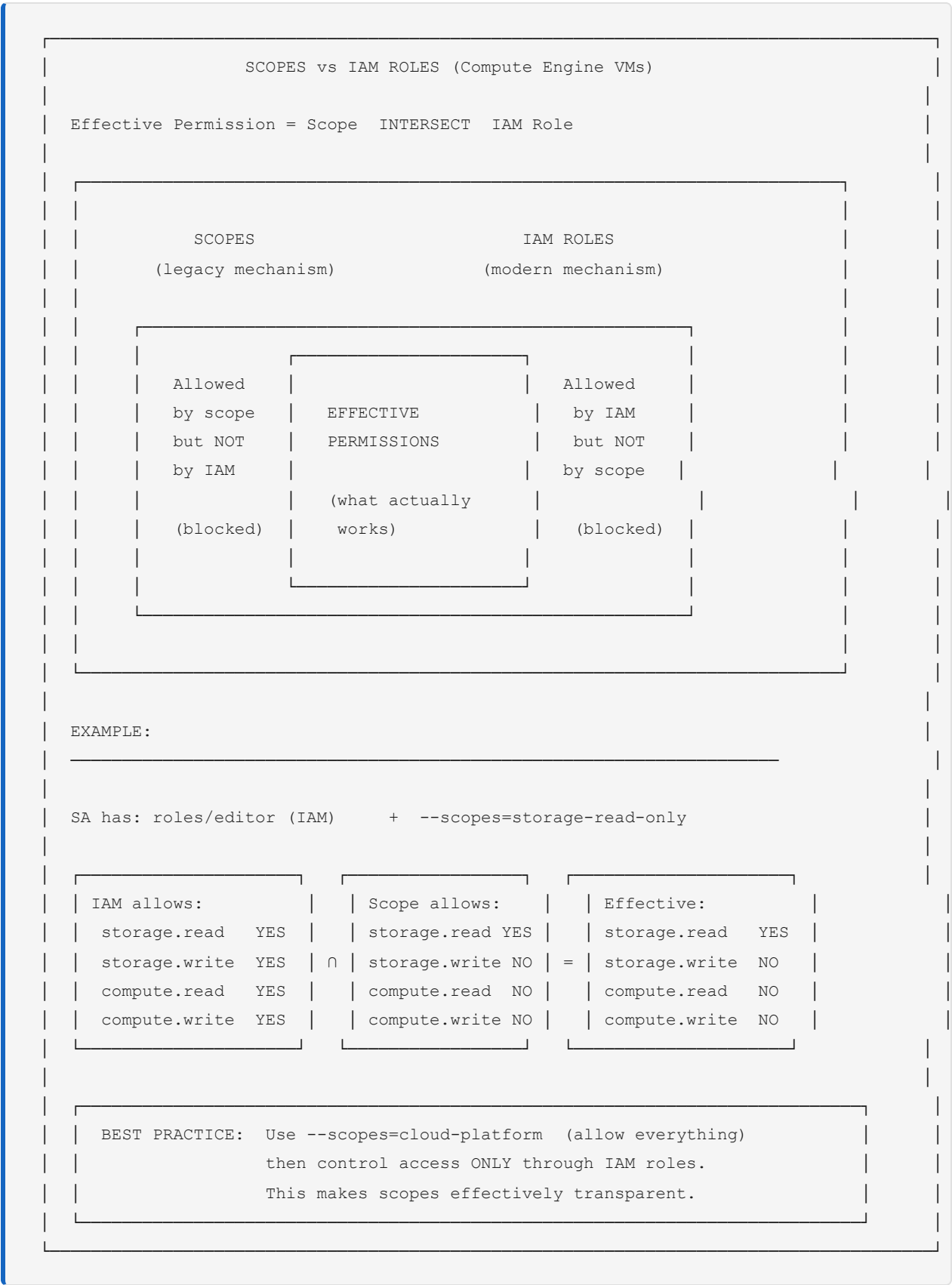


```
RESULT: Pod ► KSA ► Workload Identity ► GCP SA ► APIs (NO KEYS NEEDED)
```

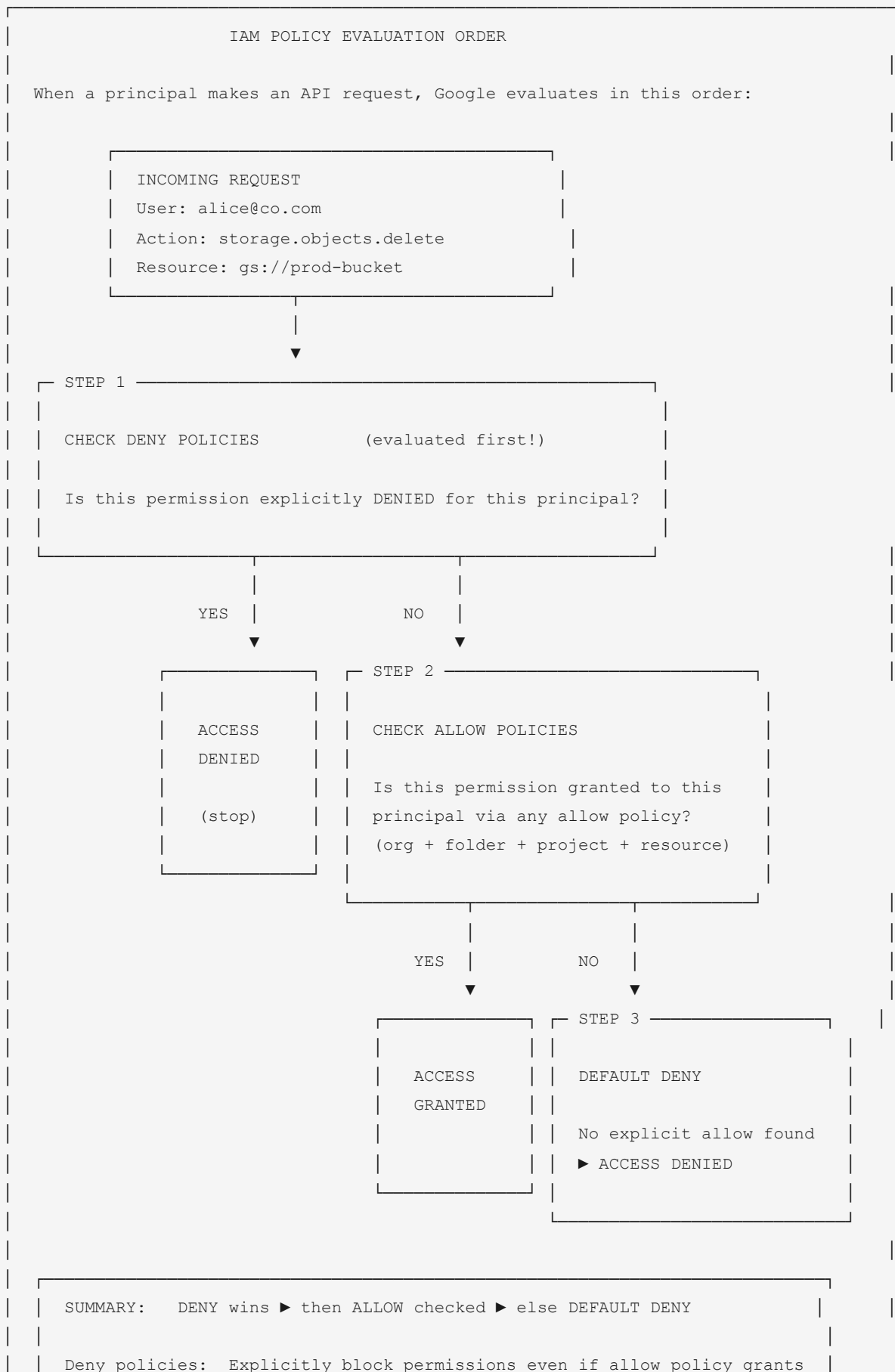
7. Service Account Impersonation Chain



8. Scopes vs IAM: Effective Permissions (Venn Diagram)

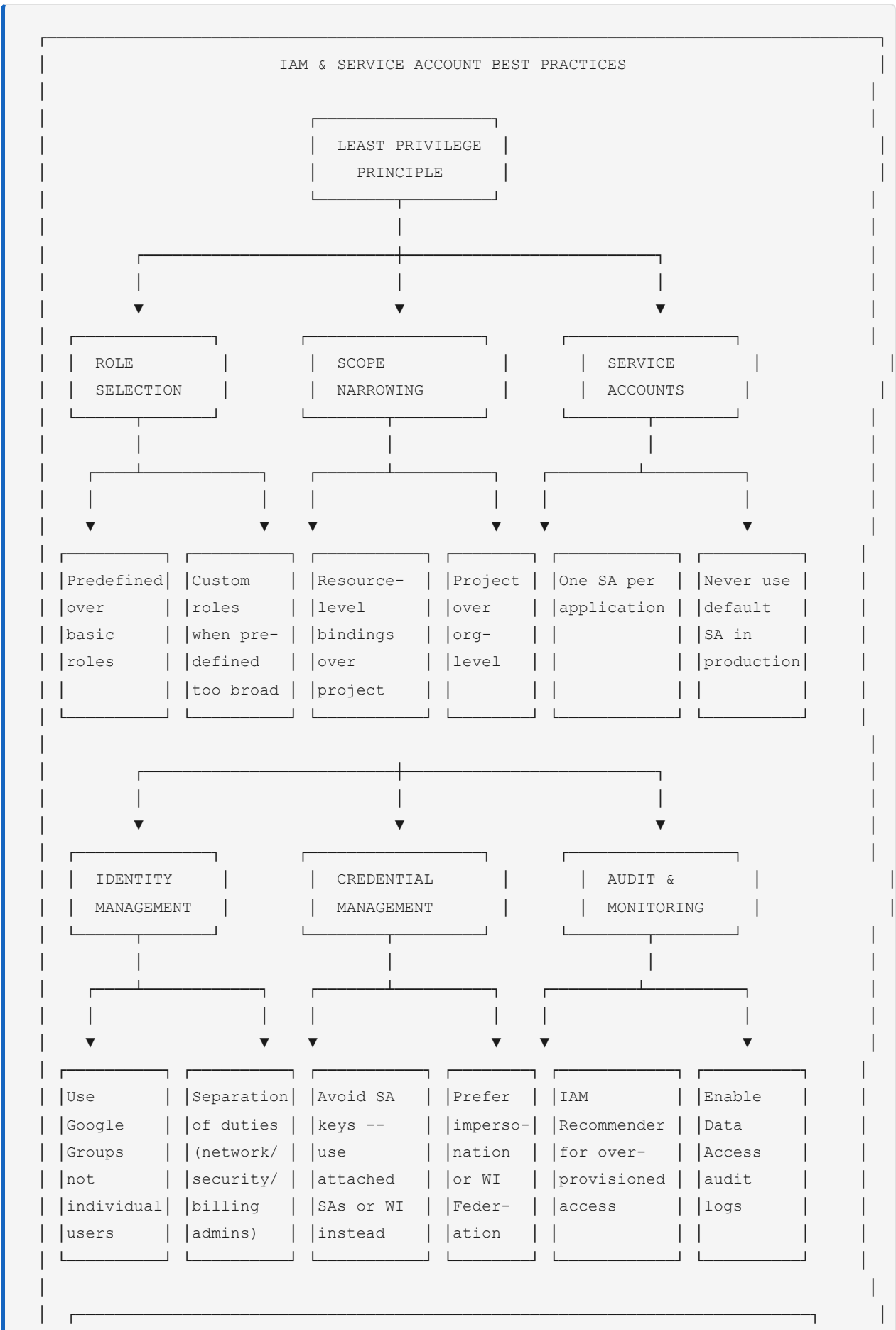


## 9. IAM Deny Policy Evaluation Order



|  |  |                                                                   |  |  |
|--|--|-------------------------------------------------------------------|--|--|
|  |  | Allow policies: Additive union of all inherited + direct bindings |  |  |
|  |  | Default deny: If nothing explicitly allows it, access is denied   |  |  |
|  |  |                                                                   |  |  |
|  |  |                                                                   |  |  |

10. Best Practices Mind Map: Least Privilege & Security



|  |  |                                                                         |  |  |
|--|--|-------------------------------------------------------------------------|--|--|
|  |  | QUICK REFERENCE COMMANDS                                                |  |  |
|  |  |                                                                         |  |  |
|  |  | Audit: gcloud projects get-iam-policy PROJECT --flatten="bindings[]"    |  |  |
|  |  | Analyze: gcloud asset analyze-iam-policy --organization=ORG             |  |  |
|  |  | Suggest: gcloud recommender recommendations list --recommender=...iam.. |  |  |
|  |  | Disable: gcloud iam service-accounts disable SA_EMAIL                   |  |  |
|  |  | Org pol: constraints/iam.disableServiceAccountKeyCreation               |  |  |
|  |  |                                                                         |  |  |